ALI RAZA

REG NO 9864

OOP

LAB NO 01

1. **Task 1**: Write a program that stores your **name, age, and GPA** in variables and prints them.

main.cpp | Share | Run | Output | Clear

```cpp
#include <iostream>
using namespace std;

int main() {
    string name = "ALI RAZA";
    int age = 21;
    float gpa = 3.7;

    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "GPA: " << gpa << endl;
    return 0;
}
```

```
Name: ALI RAZA
Age: 21
GPA: 3.7


=== Code Execution Successful ===
```

1. **Task 2**: Ask the user to enter **two integers** and print their **sum, difference, product, and division**.

main.cpp | Share | Run | Output | Clear

```cpp
#include <iostream>
using namespace std;

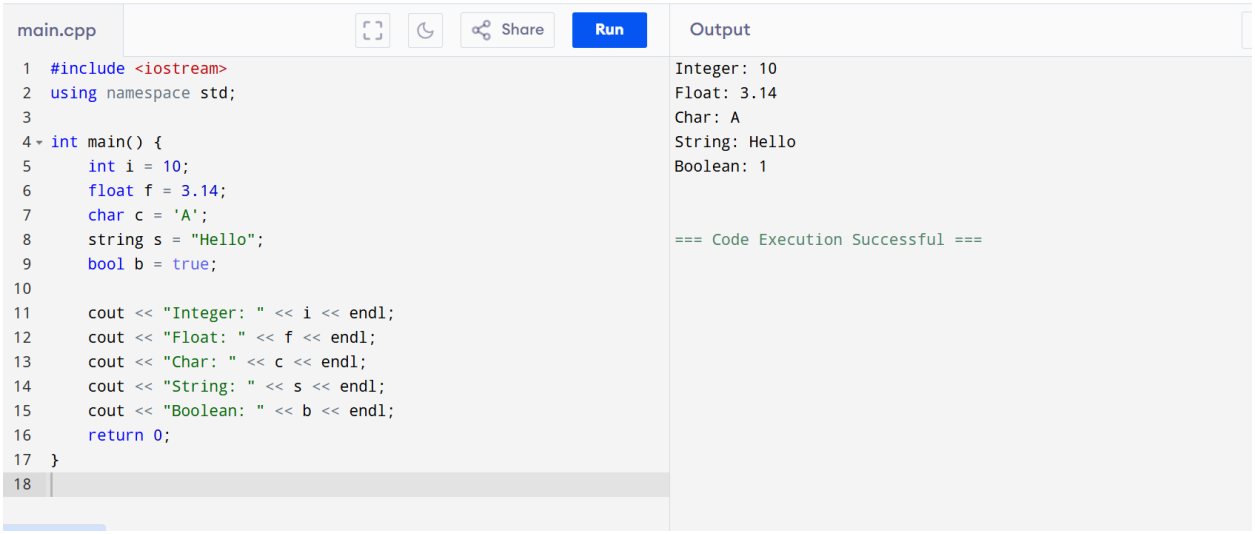
int main() {
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Sum: " << a + b << endl;
    cout << "Difference: " << a - b << endl;
    cout << "Product: " << a * b << endl;
    cout << "Division: " << (float)a / b << endl;
    return 0;
}
```

```
Enter two integers: 34
64
Sum: 98
Difference: -30
Product: 2176
Division: 0.53125


=== Code Execution Successful ===
```

Task 3: Declare variables of different **data types** (int, float, char, string, bool) and print their values.



4. **Task 4**: Print numbers from **1 to 10** using a loop.



Task 5: Print the multiplication table of 5 using a loop. .

main.cpp Share Run Output Clear

```cpp
#include <iostream>
using namespace std;

int main() {
    int num = 5;
    for (int i = 1; i <= 10; i++) {
        cout << num << " x " << i << " = " << num * i << endl;
    }
    return 0;
}
```

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Task 6: Find the sum of all even numbers from 1 to 100.

main.cpp Share Run Output

```cpp
#include <iostream>
using namespace std;

int main() {
    int sum = 0;
    for (int i = 2; i <= 100; i += 2) {
        sum += i;
    }
    cout << "Sum of even numbers from 1 to 100: " << sum << endl;
    return 0;
}
```

```
Sum of even numbers from 1 to 100: 2550


=== Code Execution Successful ===
```

. Task 7: Write a program to check whether a given number is prime or not (using loop).

main.cpp Share Run Output

```cpp
    bool isPrime = true;

    if (num <= 1) {
        isPrime = false;
    } else {
        for (int i = 2; i <= num / 2; i++) {
            if (num % i == 0) {
                isPrime = false;
                break;
            }
        }
    }

    if (isPrime)
        cout << num << " is a prime number." << endl;
    else
        cout << num << " is not a prime number." << endl;

    return 0;
}
```

```
Enter a number: 76
76 is not a prime number.


=== Code Execution Successful ===
```

Task 8: Write a function add(int a, int b) that returns the sum of two numbers.

main.cpp Share Run Output

```cpp
#include <iostream>
using namespace std;

int add(int a, int b) {
    return a + b;
}

int main() {
    int x, y;
    cout << "Enter two numbers: ";
    cin >> x >> y;

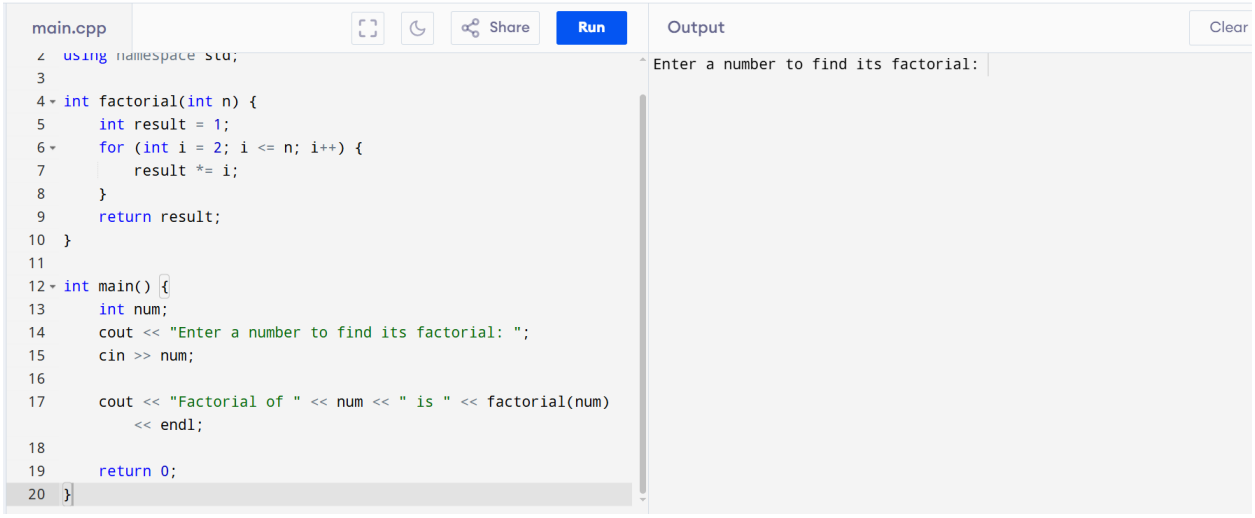
    int result = add(x, y);
    cout << "Sum: " << result << endl;

    return 0;
}
```

```
Enter two numbers: 78
54
Sum: 132


=== Code Execution Successful ===
```

Task 9: Write a function factorial(int n) that returns the factorial of a number.



Task 10: Write a function isEven(int n) that returns true if the number is even, otherwise false.

main.cpp Share Run Output

```cpp
#include <iostream>
using namespace std;

bool isEven(int n) {
    return n % 2 == 0;
}

int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;

    if (isEven(num))
        cout << num << " is Even." << endl;
    else
        cout << num << " is Odd." << endl;

    return 0;
}
```

```
Enter a number: 86
86 is Even.


=== Code Execution Successful ===
```

. Task 11: Write a function power(int base, int exponent) that calculates base^exponent (without using built-in pow).

main.cpp Share Run Output

```cpp
using namespace std;

int power(int base, int exponent) {
    int result = 1;
    for (int i = 0; i < exponent; i++) {
        result *= base;
    }
    return result;
}

int main() {
    int base, exponent;
    cout << "Enter base and exponent: ";
    cin >> base >> exponent;

    cout << base << "^" << exponent << " = " << power(base,
        exponent) << endl;

    return 0;
}
```

```
Enter base and exponent: 4
2
4^2 = 16


=== Code Execution Successful ===
```